

EDUCATION

Lahore University of Management Sciences (LUMS)

Aug 2024- 2026

M.S. Artificial Intelligence (3.65)

Relevant Courses: AI for Social Impact | Topics in LLM | Human Centered AI

COMSATS University Islamabad

B.S Electrical Engineering

PROFESSIONAL EXPERIENCE

ML Engineer & UX Researcher | Interactive Media Lab, LUMS | Sep 2025 – Present *Supervisor: Dr. Maryam Mustafa | Bill & Melinda Gates Foundation Collaboration*

- Co-building Awaaz-e-Sehat, Pakistan's first voice-first LLM maternal healthcare assistant for low-literacy women in underserved communities, and contributing to the Gates Foundation-funded National AI Hub at LUMS —Pakistan's first AI platform for Maternal, Newborn, and Child Health (MNCH)
- Conducted participatory field research and iterative usability testing with rural women in Pakistan, surfacing how collective device use, varied literacy, and cultural context fundamentally challenge standard assumptions in conversational AI design
- Designed culturally grounded, voice-first interaction paradigms for users historically excluded from the HCI design process, informing system architecture from the user up rather than the technology down
- Mentoring junior researchers and undergrad students on HCI methodology, LLM evaluation, and responsible AI design for marginalized communities

Research Intern, Tsinghua University | Summer 2025 *Supervisor: Prof. Chen Gao*

- Contributed to a large-scale multi-agent simulation framework studying how AI systems can model and predict complex social phenomena including information diffusion, relationship formation, and group coordination
- Worked on agent architecture design involving memory, reflection, and planning mechanisms that produce temporally consistent and socially believable behavior across extended multi-agent interactions
- Examined how LLM-based social simulators can serve as testbeds for understanding the societal consequences of AI deployment in community settings

Upcoming Publications

BenchMaterna: A Clinician-Validated Benchmark for Evaluating LLMs on Maternal Healthcare for Global South Women *Target: EMNLP 2026*

- Introducing the first large-scale benchmark evaluating 30+ LLMs — proprietary, open-source, small language models, and biomedical LLMs — on maternal healthcare for underserved women in the Global South
- Dataset co-constructed with 30 clinical experts who recorded real maternal health queries from women in Pakistan across regional languages, paired with clinician-validated reference answers for rigorous evaluation
- Evaluating models across six dimensions: clinical correctness, cultural relevance, health literacy accessibility, demographic fairness, harmful output detection, and helpfulness for low-literacy users in low-resource settings
- Directly addresses the absence of Global South women from existing medical NLP benchmarks, centering marginalized voices in the evaluation of healthcare AI

Awaaz-e-Sehat: Design and Deployment of a Voice-First LLM Maternal Health Assistant for the Global South *Target: CHI 2027*

- Presents system architecture, clinicians-in-the-loop design rationale, and longitudinal deployment evaluation grounded in the lived realities of marginalized women in rural Pakistan
- Investigates trust calibration and over-reliance in first-time AI users making high-stakes clinical decisions, contributing empirical grounding to HCI debates around human-AI delegation in vulnerable populations